

Adapting to climate change: biodiversity management resilience in the oil and gas sector

Summary

Climate change poses an increasing threat to biodiversity that is becoming ever more apparent. At the same time, there is an increasing pressure on businesses to become more resilient to climate change, which poses not only a physical risk but also a business risk, as all levels of society respond to these changes. Industries such as the oil and gas sector that have both a large impact and dependency on the environment are especially at risk if they do not take these long-term changes into account in their management of biodiversity and ecosystem services. This report assessed the extent to which oil and gas companies integrate climate change into their biodiversity management approaches, including impact assessments and mitigation activities. The objectives were to identify potential risks to biodiversity from climate change that are likely to be of concern to corporate biodiversity impact management, identify potential mechanisms for adaptation to climate change, examine the current corporate perception of climate change risks associated with biodiversity, and review availability of global and regional data on climate change impacts on biodiversity. A review of the literature finds that consideration of climate change in biodiversity management is currently limited, due to lack of awareness and/or prioritisation, and uncertainty of impact and management practices. Case studies of biodiversity management in the oil and gas sector also showed limited integration of climate change in impact assessment and mitigation. However, the assessment of biodiversity risk may be strengthened through the inclusion of information on biodiversity or ecosystems vulnerable to climate change. A range of data on climate change impacts and biodiversity vulnerability exists, which will inform adaptive management, but gaps still remain particularly in certain regions of the world and in marine areas, and the most appropriate data will likely depend on location and vulnerability indicators. There are many further areas in which this work could be extended, including the data review and development of guidelines for biodiversity management adaptation, which will require a coordination of expertise and disciplines to identify effective and realistic practices for resilient management.

Key Terms

(Based on CBD and IPCC Annex II Glossary of Terms)

Adaptation: Initiatives and measures to reduce the vulnerability of natural and human systems against actual or expected climate change effects.

Ecosystem based adaptation: The use of biodiversity and ecosystem services to help people adapt to the adverse effects of climate change.

Green infrastructure: Planned and managed natural and semi-natural systems that can provide more categories of benefits.

Hazard: Potential occurrence of natural/human-induced physical event that may cause health impacts, damage/loss to property, infrastructure, livelihoods, service provision, environmental resources.

Impacts: Effects on natural and human systems of physical events, disasters, and climate change.

Mitigation (of risk): Lessening of potential impacts through actions that reduce hazard, exposure and vulnerability.

Resilience: The capacity of a social-ecological system to cope with a hazardous event or disturbance, responding or reorganising in ways that maintain its essential function, identity, and structure, while also maintaining the capacity for adaptation, learning, and transformation. For ecosystems/species, a function of adaptive capacity and exposure.

Risk: Combination of hazard and its probability or severity, and vulnerability.

Risk assessment: Qualitative or quantitative scientific estimation of risks.

Risk management: Plans, actions, or policies to reduce likelihood and/or consequences of risks.

Vulnerability: Propensity to be adversely affected. For ecosystems/species, a function of adaptive capacity, sensitivity, and exposure.

Uncertainty: Degree to which a value/relationship is unknown, e.g. from lack of information or disagreement about what is known or knowable.

Methods

An understanding of the risks to biodiversity from climate change, particularly key impacts likely to be of concern to businesses, was gained by reviewing scientific papers and reports in the Convention on Biological Diversity (CBD) Technical Series. The business case for climate change adaptation and biodiversity management was identified through a variety of sources such as reports by consultancies, such as KPMG and AECOM, and sector initiatives and associations, such as the Global Oil and Gas Industry Association for Environmental and Social Issues (IPIECA) and Cross-Sector Biodiversity Initiative (CSBI). As there were few sources on the current extent of consideration of climate change risk within biodiversity management, this was inferred from analysing the reports from the initiatives, reviewing a selection of corporate sustainability reports, and through discussions with company environmental representatives at the 2016 Proteus Annual Meeting. Potential mechanisms for adaptation to climate change were collated by integrating best practices for adaptation in the conservation sector with corporate biodiversity management frameworks.

The data review compared sources of available spatial data on climate change from public datasets, international organizations, and scientific research. The most relevant data for inclusion into current tools such as the Integrated Biodiversity Assessment Tool (IBAT) were identified, primarily data on a global scale with a fine-scale resolution, and showing spatial information on biodiversity vulnerability to climate change. An overlap analysis between ecoregion climate stability, the proportion of an ecoregion predicted to have climatic conditions in the future within the current profile (Watson et al. 2013), and Proteus member oil and gas contract blocks was conducted using IHS February 2015 oil and gas data. Climate stability data did not include marine ecoregions; therefore sites with 95% or more of their area offshore were removed. The IHS contract block data also did not include blocks for the main part of the US, except for some blocks present in Alaska. The average climate stability was taken for each contract block, as even if the block overlapped more than one ecoregion, the climate stability value did not vary significantly. Case studies of biodiversity management in contract blocks with low climate stability, i.e. areas where more than 60% of the region was predicted to have a climate condition outside the current profile, were examined for evidence of climate change adaptation. Documents on corporate biodiversity management, mainly environmental policy reports and impact assessments, were also reviewed as case studies to understand the extent of consideration of climate change and adaptive actions. Based on these findings, potential areas for further work were identified.

Literature review – Climate change impacts on biodiversity, the business case, and mechanisms for adaptation

Impacts of climate change on biodiversity

It is well established that changes in climate pose a significant risk to the current state of biodiversity on all levels, regardless of uncertainty in the precise scale and mechanism of these impacts (Campbell et al. 2009). Impacts on biodiversity may include altered composition and structure of ecosystems and decreased genetic diversity due to directional selection and migration, which could in turn affect ecosystem function and resilience, lead to changes in the distribution of species or habitats, fragmentation of habitats and populations, and changes in ecological interactions, including patterns of plant and animal diseases (Bellard et al. 2012; see Figure 1).

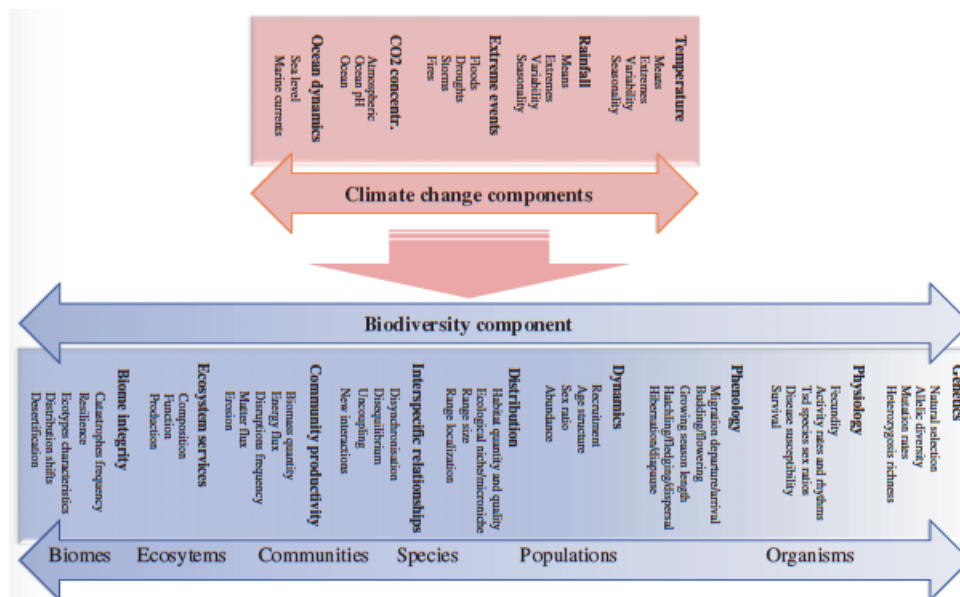


Figure 1. Predicted effects of climate change and examples of impacts on different levels of biodiversity (Bellard et al. 2012).

Even though some species may have the capacity to respond by adapting their range, physiology, and phenology, climate change will compound other stressors such as human land use change and population growth, causing changes in functioning and productivity and severe contractions and extinction of terrestrial, freshwater, and marine species (Intergovernmental Panel on Climate Change 5th Assessment Report, 2014). In the Representative Concentration Pathway 8.5 (a greenhouse gas concentration trajectory of high energy demand in absence of climate change policies which predicts a 4.3°C rise by the end of the century), a sixth of all species will be threatened by climate change (Urban 2015). Ecosystem services, such as water flow and carbon sequestration, are already likely compromised by human activity, and

biodiversity loss may also cause positive feedbacks to climate change, such as accelerating warming and forest dieback. Threshold effects, where an ecosystem reaches a “tipping point” in which change occurs rapidly and irreversibly, mean that replacing ecosystem services will become exponentially costly and challenging (Chapan et al. 2000).

The business case for resilient biodiversity management

There is already a strong business case for effective biodiversity management and climate change adaptation across all sectors as many companies depend on biodiversity and ecosystem services (BES). Examples include, but are not limited to, provision of freshwater and natural materials, spiritual and recreational services, and protection against temperature increase and erosion as well as extreme events such as flooding and water stress. Other business risks of poor biodiversity management are: regulatory risk; license to operate; brand and reputation; access to capital, land, and markets; security of supply; and liabilities (KPMG 2012, UNEP 2010).

Climate change exerts a direct and indirect pressure on these factors and can be treated just as any other risk within a company's biodiversity assessment and management plans. As biodiversity vulnerability increases and operations exacerbate the climate change impacts, local communities and shareholders will demand for better regulation or compete for a decreasing availability of ecosystem services, increasing risk on government and stakeholder relationships, as national and international regulations evolve in response to these pressures and public scrutiny increases (UNEP 2010). Project finance may also be put at risk, as the International Finance Corporation (IFC) Performance Standard 6, which defines clients' responsibilities for managing their environmental and social risks, states that “given the complexity in predicted project impacts on biodiversity and ecosystem services over the long term, the client should adopt a practice of adaptive management in which the implementation of mitigation and management measures are responsive to changing conditions and the results of monitoring throughout the project's lifecycle” (IFC 2012). Eighty-four financial institutions have already adopted the Equator Principles, a risk management framework based on IFC's environmental and social policy, and several note interest in market-based mechanisms to assess the value of ecosystem services.

At the same time, resilience in biodiversity management can provide many benefits to companies, including the ability to mitigate and better manage risks, reduced disruption to production and services, continuity of operations and social license to operate, reduced operational costs, and social and reputational advantages. It presents new opportunities, such as expansion into new markets, first mover competitive

advantage, and power to influence regulation or future policy (ICMM 2013). Investors are also known to often reward environmental stewardship (UNEP 2010). Therefore it is vital for the corporate sector to address and respond to climate change in their biodiversity management to ensure operational resilience.

Climate change risks within the oil and gas industry

Climate change risks for the oil and gas industry are rising, especially as resources yet to be exploited are increasingly located in extreme and sensitive environments. The industry operates in a wide variety of environments and habitats, including forests, wetlands, polar regions, deserts, deep-water, near shore or transition zone, grasslands, mountains, and cultivated land all of which provide key BES, and include particularly climate change vulnerable ecosystems such as coral reefs, wetlands, mangroves, cloud forests, arctic ecosystems, and freshwater habitats, which are sensitive to or increasingly exposed to climatic changes (IPIECA 2011). Assets also involve large investments and liabilities

throughout a long life cycle, usually at least 30 to 40 years from license bidding to decommissioning, and thus are exposed to long-term changes and cumulative impacts. Climate change is likely to exacerbate the sector's already large impact on the environment, with its primary and secondary footprint resulting in habitat loss and wildlife mortality, consumption of water, emissions of greenhouse gasses and pollution, and increased human immigration (IPIECA 2011).

Case Study 1 | Shell in the Nigerian Delta

The 2013 Shell Soku Gas Plant-San Barth Manifold Project is located in contract blocks with an average climate stability of the contract blocks is 0.38, i.e. about 40% of the ecoregion that the contracts are in (Central African Mangroves and Niger Delta Swamp Forest) will have the same climate profile as it does now. The full EIA was found from the Shell Nigeria Environmental Impact Assessment Reports website (SPDC 2013). Despite recognition of biodiversity and ecosystems, as well as mentioning of climate change, there does not seem to be a direct link between the two and any clear adaptation measures throughout the environmental impact assessment and management plans. Impact assessment and mitigation are primarily static measures that do not effectively consider and manage for climate change impacts, which could result in underestimation of risks and insufficient mitigation activities.

Further details can be found in Appendix 1.

Climate change impacts on biodiversity will give rise to further operational risks including subsidence, delays in exploration due to species migration, decreased tundra travel, changes in shipping lanes and seasons, damage to coastal oil and gas facilities, increased flooding, loss of cooling capacity for processing, and loss of surface water access and operational interruptions at multiple stages (IPIECA 2013). Understanding climate and biodiversity risks to neighbouring communities, even those outside company operations and facilities, is also important. These include loss of species and habitat, water stress, and greater storm impacts on key infrastructure due to weakened natural barriers and ecosystem services. The sector relies on community

infrastructure, such as electricity systems, water, transportation, communication, and residence for employees, and if the surrounding area is not resilient to climate change, it will impact company operations (IPIECA 2013). The sector also cannot ignore its dependence on a robust regional and global supply chain and infrastructure that involves sectors such as transportation, construction, electricity, and raw materials, and a close relationship with the finance sector which assesses and invests in socioeconomic and environmental resilience of oil and gas companies.

Current extent of consideration of climate change in biodiversity management in the oil and gas sector

The current extent of consideration of climate change risk in corporate biodiversity management is limited within the oil and gas sector. Despite general awareness, climate change is considered to be a long-term low to medium risk that is second to more tangible market and stakeholder risks, such as supply and demand, operational costs, and volatile prices, suggesting a lack of understanding of their interdependence. Climate is primarily considered as an operational risk, with a focus on extreme events more than gradual slow-onset events, even though the latter events will become significant within the lifetime of the company's operations (Agrawala 2011).

Case Study 2 | Eni in the Nigerian Delta

The Nimbe Field is located within Nigerian Agip Oil Co Ltd (NAOC)'s OML 61 in Nembe Local Government Area of Bayelsa State, Nigeria, with an average climate stability of 0.033. The Executive Summary for the Nimbe Field Further Development Project ESHIA was found from the eni NAOC Sustainability website (NAOC 2008). Little to no allusions are made to changing climate or possible adaptation measures. Despite mentioning the importance of water to communities and ecosystems, measures for a net-neutral or positive effect are not in place, neither is altered hydrology due to climate change a consideration, to better minimise the impact from reinstating or altering natural drainage patterns and watercourses. Instead of building dykes to safeguard areas prone to erosion due to dredging, Eni may consider a more adaptable, perhaps ecosystem-based measure to water level rise such as mangrove planting. A training and awareness programme throughout the project lifecycle for staff, contractors, and the host community by the monitoring team including climate change adaptation, would benefit both the community and conservation efforts to adapt to issues such as increased water stress and changes in ecosystems.

Further details can be found in Appendix 2.

The process of risk and adaptation assessments is not standardised across the oil and gas industry and largely conducted on a project-by-project basis (AECOM 2016). This may be due to the differences in geographic and local conditions of each project, fragmented climate change adaptation regulation, policy, and assessment processes, and lack of expertise, capacity, or incentives. Although the climate risks assessed correspond with the perceived risks to operations, the assessments of risks and impacts are varied across operators, for example with only a few assessing disease vectors and species migration. Many risks that have an impact on both operations and biodiversity tend to not be universally assessed. For example, risks such as disease

vectors, ocean pH, and species migration are considered particularly low suggesting that impacts on biodiversity may have little influence in the design and operation of projects. Adaptation assessment methods are also varied, with ecosystem services, involvement of external experts, use of future projection data, and development of adaptation actions often not included in the literature and responses in reports reviewed. This suggests a lack of integration between environmental and climate change experts and corporate operations, leading to lack of proper adaptation measures in environmental management. These observations also highlight the role of uncertainty in prioritisation of risk management, which may cause difficulty in measuring and quantifying risk, return on investment, and cost-benefit that are ultimately the main priorities of businesses in designing strategies (Acclimatise 2009).

Case Study 3 | BP in the Alaskan Arctic

BP Exploration (Alaska) Inc. currently has operations in the arctic coastal tundra of Alaska including Prudhoe Bay, which includes one end of the Trans Alaska Pipeline System (TAPS). This arctic coastal tundra ecoregion has a climate stability of 0.079. The Alaska report mentions a wildlife monitoring programme to better understand operational effects on wildlife populations, which could be undertaken from an additional climate change perspective, for example to monitor changing permafrost conditions, vegetation or migratory paths. The Seismic Survey Environmental Assessment appendix includes a section on climate change consideration in evaluating cumulative effects on environmental, social, and economic resources of the Arctic, particularly ocean acidification. However, what is considered is whether the operation is an additional cause of climate change or whether it is part of the cumulative impact on the environment, rather than the effect of climate change on the operation or how environmental management can actively adapt to minimise these impacts. The thirteen team members consisted primarily of biologists but also included one meteorologist who contributed information on climate change. Including personnel with expertise on climate change in arctic and marine ecosystems projects could further improve the team composition, and foster collaboration between the environmental and engineering aspects of the project.

Further details can be found in Appendix 3.

There is some evidence that national regulation requiring consideration of climate change risks can increase the scope for biodiversity management adaptation. For example, since 2010, United States securities laws require companies to disclose the risks related to climate change that could materially impact the company, including the results of operations and direct or indirect impacts of regulation, business trends, and physical consequences related to climate change in annual and quarterly reports filed with the Securities and Exchange Commission (SEC). This has prompted Chevron to address managing climate risk, such as utilizing risk management systems to assess potential weather-related and physical risks to assets and ensure resiliency, as well as capital investment reviews and decisions that involve uncertainty analyses, which incorporates potential ranges of factors such as storm severity and frequency, air and water temperature, precipitation, and fresh water access (Chevron 2001-2016).

Reflections on Proteus Annual Meeting 2016

In questioning the extent of climate change risk consideration in biodiversity management, the informal poll conducted at the 2016 Proteus Annual Meeting suggested it was currently low to medium, but an outlook for medium to high in the future. One partner mentioned the need to change from a reactive to active approach in adaptation of biodiversity management strategies, with another mentioning the tendency to focus on the current state of climate due to the difficulty in predicting future impact on biodiversity services. Presently, there also appears to be a greater focus on climate change in terms of mitigation, for example in emission reduction, rather than adaptation of management to climate change. 'Adaptation' is primarily in engineering design of grey infrastructure, perhaps due to the more tangible and quantifiable nature of the risk. Other drivers and barriers to adaptation highlighted were finances, government or regional policy, information gaps and risk uncertainty, and climate change viewed more as a mitigation issue rather than an adaptation goal.

These observations highlight the need to consider resilience of ecosystems and species to climate change in company adaptation strategies, as well as the need to include climate change in long-term investment choices and business models of biodiversity management. A tailored approach for high risk, vulnerable areas, such as water and material risks, should be identified while considering the context of each project. Climate change perspectives need to shift from a focus purely on emissions to one that includes adaptation. This may require better integration and collaboration between climate, environmental, and engineering teams. Such efforts may also help reduce the barriers to climate change adaptation by enabling identification of hybrid solutions with co-benefits and sharing of relevant data and knowledge across teams, such as climate change information used for operation risk assessment also being incorporated into environmental assessment and management.

Mechanisms for adaptation

Adapting existing management practices

There are several mechanisms for climate change adaptation in biodiversity management. The most practical and perhaps effective way is to evaluate and adapt existing biodiversity management plans to climate change by evaluating likely outcomes under the current plans, rather than introducing a new system or "reinventing the wheel." Initiatives such as IPIECA and CSBI have developed a number of resources and guidance reports used by the oil and gas sector, including several Proteus members like BP, Chevron, and Shell (IPIECA 2005). This includes the CSBI

mitigation hierarchy, which can be used in conjunction with the initial stages of the IPIECA Biodiversity Action Plan (BAP) guide. These can build on and be integrated with existing company activities and processes throughout a project's life cycle such as Environmental and Social Impact Assessments (EIA or ESIA) or Environmental Management Programmes (EMP). Climate change and how it interacts with biodiversity should constantly be considered alongside these processes – for example, long term net-loss or gain could be measured in the mitigation hierarchy, and in the BAP chart in Figure 2 below, both the current and future climatic conditions could be incorporated into the biodiversity assessment survey, identification of impacts, action plan development, and monitoring and improvement.

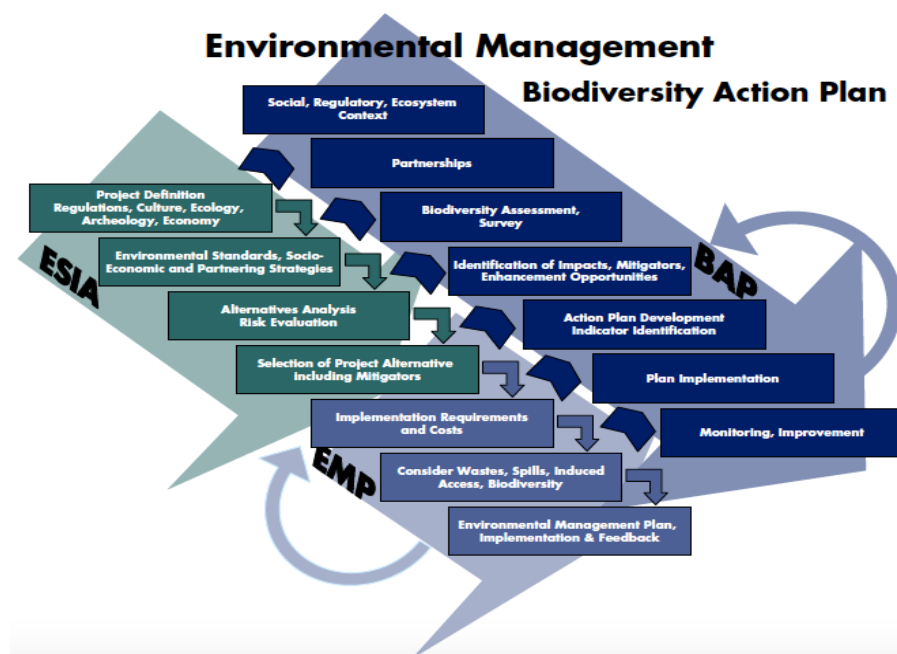


Figure 2. IPIECA BAP. This may be used in conjunction or integrated with the company's ESIA and EMP (IPIECA 2005).

Depending on the corporate management structure, incorporating biodiversity in climate change risk assessment and adaptation plans would be an alternative way of adapting existing management practices. For example, the International Council for Mining and Metals (ICMM) has produced a report on climate change risk assessment and adaptation, which includes biodiversity impacts and notes the possibility of applying the adaptation methods in biodiversity management (ICMM 2013). These methods that involve evaluating and creating targets and monitoring progress that considers total biodiversity impacts inclusive of climate change are applicable to closely related extractive mining and energy sectors.

Adaptive management: learning from the conservation community

Many lessons can also be learned from the conservation community in terms of best practices for adaptation mechanisms. Corporate environmental management teams can adopt adaptive management tools, which usually follow the framework, adapted from a report by the Cambridge Conservation Initiative (CCI) and non-CCI partners (CCI et al. 2010, Belle 2016):

1. Identifying biological features targeted for conservation (species, ecological processes, ecosystems) and gathering explicit, measurable, management objectives for each
2. Reviewing existing goals and objectives by building a conceptual model to illustrate climatic drivers, in addition to ecological and socio-economic drivers, affecting each feature
3. Assessing vulnerability to climate change and identifying intervention points and potential adaptation actions to achieve objectives under each climate scenario
4. Evaluating feasibility of and trade-offs among potential actions
5. Developing and implementing an action plan
6. Monitoring the efficacy of actions toward objectives to identify any needs to revisit planning and continue to build capacity for adaptation to climate change

Different strategies can be considered depending on the existing condition of the ecosystem and species, and their vulnerability to climate change. Specific actions may include maintaining and increasing habitat connectivity by using corridors and buffer zones with consideration of the greater landscape, such as surrounding protected areas and areas of High Conservation Value (HCV). This would facilitate the movement of species and any autonomous adaptation to climate change, and monitor sites for artificially translocated species or those undergoing range changes due to extinction risk or their invasive nature (Campbell 2009). By engineers and environmental teams working together, these needs would hopefully be taken into account in operation design and environmental management.

Identifying opportunities for multiple wins

Companies can also identify synergistic opportunities to address ecological impacts on biodiversity, social impacts on human communities, and operational risk. An improved understanding of co-benefits and hybrid solutions, such as ecosystem based adaptation (EbA) and green-grey infrastructure, across all levels of management will allow better integration of engineering and operational needs within an ecosystem approach. Investment in services that enable biodiversity management and climate change adaptation, including ecosystems, watersheds, and natural resources, shared knowledge and technical resources, and public infrastructure and community assets would also increase the resilience of the natural environment to operations (UN Global Compact 2015). Engaging and incorporating all stakeholders, including governments and communities, could result in strategies for long-term resilience and reduction of conflict with local communities (UN Global Compact 2012). Resources for designing the engagement process have been developed by organisations such as UNEP-WCMC, for example one for PA resilience (Wicander et al. 2016), and such tools can be adapted for each project or EIA. Stakeholders with traditional or indigenous knowledge can help identify key climate change and biodiversity issues early in the process. Conservation training, including climate change adaptation at

multiple institutional levels and from regional to national scales, may further build lasting positive impact.

A spectrum of approaches

Another possible consideration for climate change adaptation is the resistance vs. resilience approach (Figure 3). While the examples are intended for conservation management, these measures can be applied to corporate biodiversity management. For example the “limit land purchases to future hotspots” can be interpreted as considering future likely protected areas in applying for concessions or building infrastructure. Resistance would attempt to bolster defences to rapid environmental change, such as increasing irrigation with decline in precipitation, while resilience would attempt to absorb change, such as facilitating a population’s adaptive capacity by introducing a wider range of genotypes. While resilience may be the ideal goal, a complete strategy should consider the whole spectrum, considering the cost effectiveness of adaptation options and compatibility with existing regulator and institutional constraints (Heller 2009). This also includes consideration of spatial and temporal data and its resolution used to inform perspectives. For example, repopulated plant species could be determined by future climate conditions rather than historical presence with suitable projections.

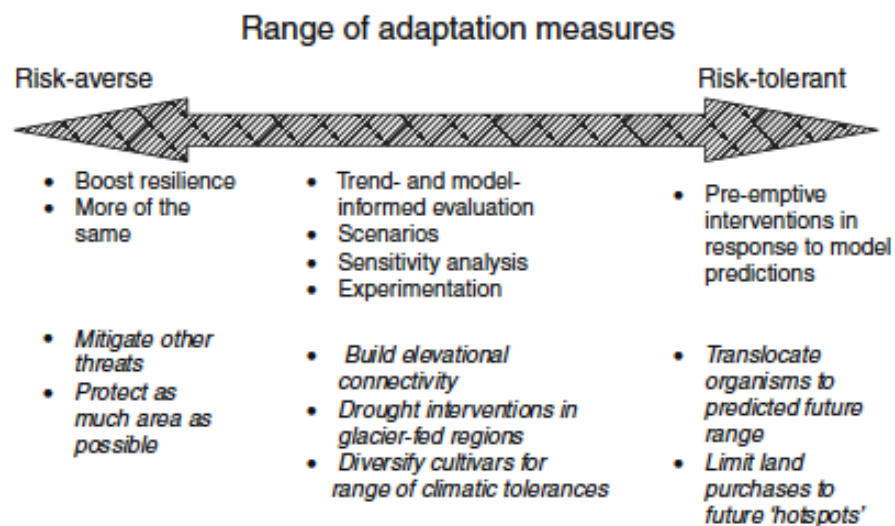


Figure 3. A risk-continuum of adaptation measures that shows examples of general approaches of adaptation to risk followed by specific measures (Heller 2009).

Dealing with uncertainty

Uncertainty in projecting future climate change and identifying risks can be overcome by exploring a range of projected scenarios, for example business-as-usual or worst case scenarios, to formulate management plans (CCI et al. 2010). In climate risk assessments, the location, type, and design of the operation or facility, project lifetime, current environmental baseline conditions, historical and current climate variability,

and projected change in climate and environmental conditions and its rate will be key in identifying adaptation plans unique to the local area (IPIECA 2013). Quantification of natural assets by, for example calculating natural capital and cost-benefit analysis or cost-effectiveness of measures under different scenarios of climate change, would also decrease the uncertainty surrounding financial costs and aid in measuring outcomes (UNFCCC 2011). IPIECA provides an ecosystem services guidance document with extensive checklists of risks and opportunities for most relevant habitats occupied by oil and gas operations. This can act as a basis for natural capital assessments and risk management and impact assessment procedures. Although extreme, anomalous events will become more frequent as climate change continues, adaptive management can increase the coping range and critical threshold of climate variables for operations and biodiversity in the area. The agricultural sector has led the way in climate change adaptation, using mathematical models to integrate long-term demographic climate data to set climate change appropriate harvest schedules or forecast pest outbreaks, and mapping site-specific climate conditions and biotic responses to prioritize locations for farm diversification (Heller 2009). These mechanisms could be applied to the oil and gas sector in assessing the risks and impacts to biodiversity and adapting management accordingly. For example, combining local climate, environmental, and demographic data collected in environmental assessments with climate change projections may enable identification of vulnerable species and habitats and prioritise conservation efforts.

Data review

Analysing climate change vulnerability will inform adaptation actions and priority areas by allowing better evaluation of risks and identification of management practices accordingly. Mapping the spatial distribution of ecosystem and species vulnerability may be a valuable tool for oil and gas companies to identify where operations overlap with vulnerable areas, and thus adapt their biodiversity management accordingly. While raw current global climate and bioclimatic data and projected future data using IPCC scenarios are available, mapping the impacts on BES or species vulnerabilities is a more complex area. One way of measuring vulnerability is as a function of exposure, sensitivity, and adaptive capacity. An example is the Climate Change Vulnerability Index for habitats and ecosystems (HCCVI) by Natureserve (Figure 4). There are also many other measures for assessing climate change vulnerability, which are usually correlative, mechanistic, trait-based, or a combination of these. The most useful data will depend on conservation goals and data available for the species or ecosystems in question (Pacifi et al. 2015).

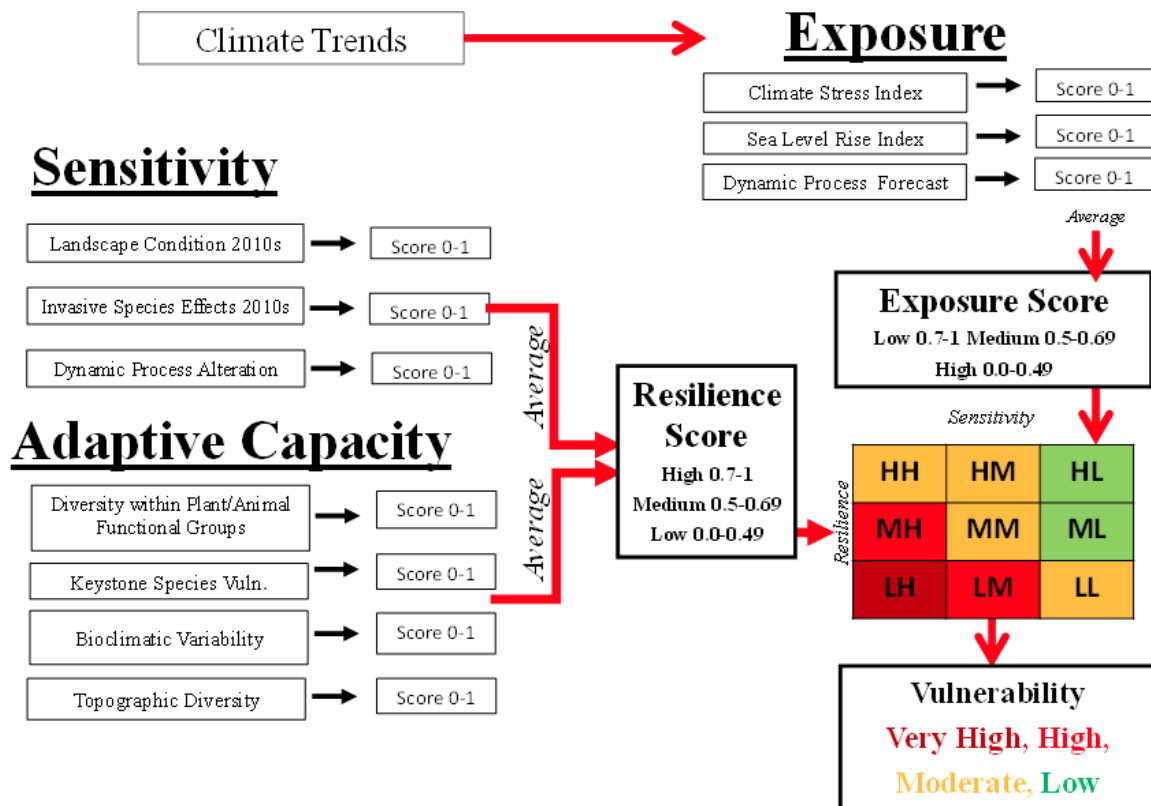


Figure 4. Vulnerability measured as a function of Exposure, Sensitivity, and Adaptive Capacity. Each component has 3-5 indicators, each requiring a separate type of analysis, which produces sub-scores used to generate an overall score for exposure vs. resilience (sensitivity + adaptive capacity). (Young 2015)

Availability of climate change data related to biodiversity impacts

A table of most relevant data sources found in the data review – those mapping a spatial global distribution of a measure of climate change vulnerability – are shown in Table 1. These data sources are modelled using input from typically open-access, raw data sources such as WorldClim, GlobCover, and SeaAroundUs. Further information on each data source, including limitations, can be found in the Supplementary Excel file in Appendix 4, in addition to some raw climate and regional data sources.

Table 1. Biodiversity vulnerability to climate change data sources. Green indicates terrestrial maps, blue marine maps, and cyan indicates terrestrial and marine.

	Data source	Age	Resolution	Components	Objective
Climate stability and habitat intactness (Watson et al. 2013)	GlobCover v2.1; WorldClim; International Centre of Tropical Agriculture Downscaling data set	2013	0.3-4km	Ecoregional climate stability; Vegetation intactness; and their relationship	To map relationship between ecoregional intactness and stability across ecoregions to identify vulnerability and conservation adaptation strategies under climate change
Current & future biodiversity vulnerability to habitat loss-CC interactions (Segan et al. 2015)	GlobCover; WorldClim	2015	1km	Likelihood that observed climate change has exacerbated impact of habitat loss and fragmentation of biodiversity at ecoregional scale	To assess of current and potential future impacts of habitat loss-climate change (HLF-CC) interaction on biodiversity using changes in temperature and precipitation, and allow identification of areas where avoidance of HLF-CC interaction magnify benefit of prevented loss or restoration to biodiversity to allows prioritization of actions and evaluating effectiveness
Trait-Based Vulnerability Assessment of all Birds, Amphibians and Corals (Foden et al. 2013)	IUCN Red List; WorldClim; Global Land Cover 2000; ReefBase	2013	19km	Number and concentration of climate change vulnerable species (birds, amphibians, corals)	To assess sensitivity, exposure, and adaptive capacity using species' biological traits and modelled exposure to projected climate change to map climate change vulnerability of species and identify priority areas for conservation
Assessing species vulnerability to climate change (Pacifici et al. 2015)	review of 97 studies	2015	?	Ecoregional global concentrations of terrestrial and marine climate change vulnerable species Predicted distribution of biodiversity impact due to warming-induced range shifts in marine metazoans in terms of invasion intensity, local extinction intensity, and species turnover in 2050 relative to mean of 2001-2005	To map ecoregional global concentrations of terrestrial and marine climate change vulnerable species
Marine biodiversity range shift impacts in terms of invasion intensity, local extinction intensity, and species turnover (Cheung et al. 2009)	SeaAroundUs; FishBase; SeaLifeBase	2009	55km	Hotspots of local extinction & invasion intensity of 20% of more between 2000 and 2050, with shading showing areas of high (3 models) and moderate (2 models) agreement	To investigate the global patterns of climate change impacts by projecting the distributional ranges of a sample of 1066 exploited marine fish and invertebrates for 2050 using dynamic bioclimate envelope model
Multi-model ensemble projections of climate change effects on global marine biodiversity (Jones 2014)	SDMs (AquaMaps, Mazent, Dynamic Bioclimate Envelope Model)	2014	55km	Average Vulnerability Index for RCP 2.6 and 8.5 (based on sensitivity, adaptive capacity, and exposure)	To assess global patterns of change in species richness, invasion, and extinction intensity in the world oceans
Integrated Average Vulnerability index (Jones 2015)	AquaMaps, Mazent, DBEM	2015	?		To assess the vulnerability of marine fish and invertebrates to climate change

According to a review of 97 studies published between 1996 and 2014, with more than 70% during the last five years, and only 4% assessed species' vulnerability globally. The studies reviewed were used to derive a global map, which is a useful visual overview of climate vulnerable areas (Figure 5). More than 60% of the studies are at a local scale, which may only assess part of a species' range and result in unrealistic estimates of vulnerability (Pacifiçi et al. 2015). This also suggests the limit in existing global data or studies on climate change vulnerability of biodiversity. Moreover, only a few marine taxa have been evaluated at large scales. There is bias in the distribution of regional studies conducted, with fewer studies available for tropical and subtropical regions, often the most biodiverse areas.

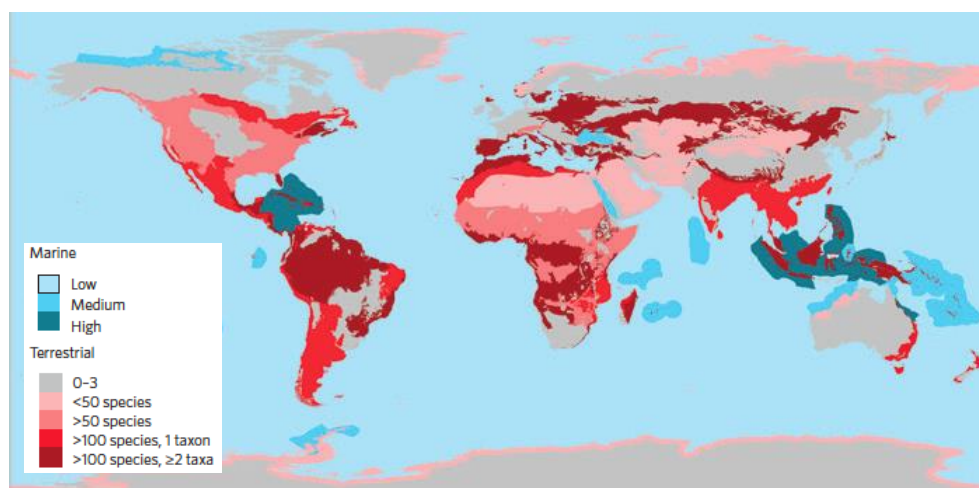


Figure 5. Global concentrations of terrestrial and marine climate change vulnerable species according to ecoregional classification, derived from studies conducted at regional, continental and global scales (Pacifiçi et al. 2015)

Ecoregion climate stability and contract block overlap analysis

An overlap analysis was conducted using data from the Watson 2013 paper on ecoregion climate stability and oil and gas contract block data. Figure 7 shows the results of this analysis. Although climate stability is an incomplete measure of vulnerability, this serves to illustrate the potential use of a global climate change ecosystem vulnerability map. Some of the least climate stable regions with Proteus member contract blocks were concentrated in Western Russia, Canada, the Nigerian coast, and Alaska, with all having climate stability value of less than 10%. This may be because oil and gas sources are in some of the regions most exposed to climate change. Almost all of the Proteus oil and gas partners operate in these areas, suggesting that considering climate change is relevant and imperative. Overall, BP had the lowest average contract block climate stability while eni had the highest. It is unclear whether these patterns are due to variation in climate change consideration in corporate or operational management, or reflect other regional corporate strategies.

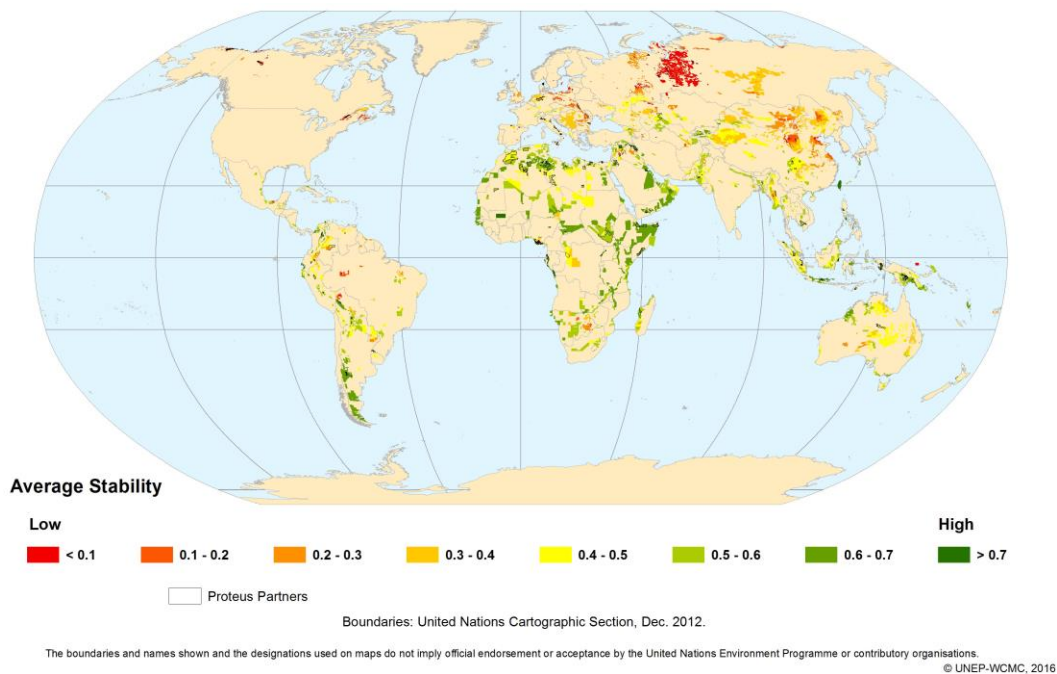


Figure 6. Average climate stability of Proteus Partner contract blocks (S. Blyth 2016)

Areas for future analysis

Climate change, biodiversity, and their interplay are complex topics that are compounded by the myriad of management systems and adaptation mechanisms both in conservation and in business. However it is important to mitigate biodiversity and adapt management according to future changes. This complexity highlights the need for collaboration across areas of expertise within UNEP-WCMC and the wider conservation and corporate environmental management arena. Some specific areas for future analysis and research may include:

Continued improvement in climate change vulnerability data, modelling, and mapping

Continuation of efforts such as Protected Areas Resilient to Climate Change (PARCC) for all PAs, KBAs, and High Conservation Value Areas (HCCVAs) would provide information on IUCN status; exposure (in 30 years, 50 years etc.), sensitivity to climate change, adaptability to climate change, and adaptation action. These efforts can be prioritised for areas with ecosystems or species most vulnerable to climate change. This may enable additional layers to be included in IBAT related to vulnerability of biodiversity to climate change such that companies can integrate this spatial information into their management plans and reduce uncertainty, a key barrier to active adaptation. The framework in the Supplementary Excel File in Appendix 4

shows a potential way of identifying gaps in data and literature on a regional and global scale. Improvement of knowledge can be achieved in collaboration with Climate Change and Marine teams at UNEP-WCMC, as well as research partners such as authors of the indicated papers in Table 1.

Site-specific data collection and mapping

More detailed site-specific data collection and mapping of climate change vulnerability would be especially useful for oil and gas projects in order to identify context-specific mitigation and adaptation actions. Information on specific aspects of sensitivity or adaptive capacities of species or ecosystems would help to address and target approaches. Knowledge sharing of EIAs or climate change impact examples between companies would also increase availability of local data and information.

Consideration of biodiversity management resilience related to secondary impacts (e.g. roads) and renewable energy

While this report focuses on primary impacts of oil and gas operations, impacts from other activities, such as roads and/or renewable energy, should be considered as they are also significant. Currently there is a UNEP-WCMC joint Science - Business and Biodiversity Programme concept note to model secondary impacts of the oil and gas sector, which may aid in identifying areas that need to be managed for climate change vulnerability and finding the best approach for mitigating impacts. Many oil and gas companies do not deem themselves responsible for these impacts, but this model may also elucidate their effects and support measures to hold companies accountable.

Standardisation of guidelines on adaptation based on climate change vulnerability

Guidelines for businesses outlining suitable management practices and including the vulnerability of species or ecosystems would aid in increasing consistency in environmental management and incorporation of climate change in planning. One potential framework may be a vulnerability adaptation action matrix similar to a risk matrix (Young 2015, Watson 2013). These frameworks could be refined to be more relevant to operations by consulting industry initiatives as well as environmental and engineering teams in companies.

Conclusions

Efforts to integrate climate change adaptation into core business strategies are increasing for all industries, with the oil and gas sector being one of the most vulnerable due to the nature of its operations and impact on the environment. However, current awareness and prioritization of climate change risks is still low based on the literature review. The next step for the oil and gas sector is to extend climate change into all planning and action by integration into environmental management, and emphasising practices for adaptive management within operations to manage risks and impacts. All adaptation strategies need to be flexible and holistic in the face of unavoidable uncertainty. Due to the complexity of ecosystems and species and the feedback effects of climate change, resilience must include the willingness to change course, the capacity to monitor and reassess conditions frequently, and understanding that risk-taking includes lack of action. Incorporation of long-term impacts and projections, in addition to historical data and events, would enable adaptive biodiversity risk assessment and management, and would complement the relatively long lifetime of oil and gas projects. There is global and regional data available on ecosystem and species vulnerability to climate change. The overlap analysis with oil and gas contract blocks revealed a potential application using climate change data to identify areas of high vulnerability and determine the most effective adaptation measures. However there are still gaps in data available and the suitability of use will depend on the location or context. High quality and quantity of relevant spatial and temporal data, as well as continued accumulation of pilot studies and leadership examples by the scientific and corporate community, will be highly beneficial in expanding the evidence-base for the oil and gas sector.

Appendices

Case Studies

Based on the overlap analysis between climate stability and oil and gas contract blocks, companies and their operations in regions of low climate stability were chosen as case studies. These were examined for any evidence of climate change adaptation and potential suggestions for improvement. Whilst direct comparison between the case studies is difficult due to differences in ages, project type, documentation availability, and approaches in environmental management, they all lacked effective integration of climate change in their assessments and impact mitigation.

Appendix 1. Shell 2013 Soku Gas Plant-San Barth Manifold Project, Niger Delta

The Soku Gas Plant is located in the Shell Petroleum Development Co of Nigeria Ltd (SPDC)'s OML 23 and 25 in the Akuku Toru Local Government Area in Rivers State to Nembe Local Government Area in Bayelsa State in the Niger Delta. The average climate stability of these blocks is 0.38, i.e. about 40% of the ecoregion that the contracts are in (Central African Mangroves and Niger Delta Swamp Forest) will have the same climate profile as it does now. The full EIA was found from the Shell Nigeria Environmental Impact Assessment Reports website (SPDC 2013). While climate change is referenced, impact assessment and mitigation are primarily static measures that do not explicitly manage for climate change impacts. Shell recognises the mangrove ecosystem and its ecological services (stabilization of the swamp and provision of habitat) and their root sensitivity, as well as their highly adapted nature to a brackish environment. They also recognise the definition of climate change according to IPCC, emphasizing the long-term scale and degree of variability, duration, and impact on humans and ecosystems. They note International Laws and Regulations of which Nigeria is signatory to, including the UN Framework Convention on Climate Change ratified in 1994 and the Nairobi Convention on Biological Diversity. Shell appear to recognise the need for climate change adaptation in its reference section which includes an entry for 'Climate Change and Human Health: Impact and Adaptation' (Kovat 2000) but it is not referenced in the body of the EIA. Finally, it records fears of the Soku community such as "degenerative effects on the aquatic ecosystem, reduction in number of endangered species" as well as land degradation and water pollution highlighted by others in the community.

Despite recognition of biodiversity and ecosystems, as well as mentioning of climate change, there does not seem to be a direct link between the two and any clear adaptation measures throughout the environmental impact assessment and management plans. While ecosystems appear to be well studied, there is no account

as to how climate change might affect them in the future. The prediction and evaluation of impacts, based on the ISO 14001, lacks much consideration of climate change. For example, the frequency of occurrence of each impact were determined only from historical records, despite the fact that extreme events are predicted to increase due to climate change. It is unclear how the probability of risk occurrence is evaluated and whether project lifespan includes the total lifetime of, for example pipelines, where long term changes in climate may accelerate erosion, contaminate water, or destroy habitats. Impacts, such as the destruction of mangroves, which are likely to be lost at least through the lifetime of a pipeline, are worrying for an ecosystem already vulnerable to climate change, in addition to the increase in human intrusion. Mitigation measures do not take climate change into account, in neither the proposed actions nor their evaluation. Actions such as limiting clearing areas or avoiding natural drainage channels are static measures that do not consider a changing landscape. Evaluations of impacts as 'reversible' or impacting rating post-mitigation as 'low' even when the initial rating was 'high' for impacts such as flooding or loss of wildlife habitat, appears underestimated in light of climate change impacts. Some actions, such as providing wildlife corridors and fauna crossing in response to habitat fragmentation, or re-vegetation of exposed areas could potentially be adaptive, but are not yet framed in such a way and/or do not explicitly take account of climatic changes. Finally, the Environmental Management Plan, does not include monitoring for impacts on biodiversity, which will be increasingly exposed to climate change.

Appendix 2. eni 2008 Nimbe Field Further Development Project, Niger Delta

The Nimbe Field is located within Nigerian Agip Oil Co Ltd (NAOC)'s OML 61 in Nembe Local Government Area of Bayelsa State, Nigeria. The average climate stability of OML 61 is 0.033, i.e. only around 3% of the ecoregion that the contract is in (Niger Delta Swamp Forest) will have the same climate profile as it does now. The Executive Summary for the Nimbe Field Further Development Project ESHIA was found from the eni NAOC Sustainability website (NAOC 2008). Eni acquired baseline environmental conditions as a part of its EIA. It recorded mangrove and rainforest vegetation, as well as several endangered and vulnerable birds and mammals. Consultations with stakeholders included representatives from host communities, federal and state government regulators, and an EIA consultant.

The lack of any references to climate change or adaptation of management in the ESHIA executive summary, despite relatively detailed accounts on socio-economic and environmental conditions of the community, suggests adaptive management is not a priority. The section on impacts and mitigation measures focused on minimisation or restoration based on unchanging conditions. Measures such as allowing regrowth of native plant species or compensation to owners of economic trees do not consider that

these may no longer occur in that region in the future. Most measures focus on flora and less on fauna, which are also a key part of the degraded ecosystem. Rather than simply promote use of synthetic medicines over forest-based medicinal plants or synthetic fertilisers to improve soil fertility, Eni could encourage better understanding of these natural features and how they might be affected by climate change. Despite mentioning the importance of water to communities and ecosystems, measures for a net-neutral or positive effect are not in place, neither is altered hydrology due to climate change a consideration, to better minimise the impact from reinstating or altering natural drainage patterns and watercourses. Instead of building dykes to safeguard areas prone to erosion due to dredging, Eni may consider a more adaptable, perhaps ecosystem-based measure to water level rise such as mangrove planting. Offsetting fragmentation and alteration of ecosystems in operations by increasing connectivity in other areas or monitoring for invasive species would be a more resilient measure compared to the indirect human safety measures already listed. An environmental monitoring team would be beneficial during the entire lifecycle of the project, instead of only after decommissioning, to monitor for environmental changes and trends to adapt the EMP to any observations. A training and awareness programme for staff, contractors, and the host community by the monitoring team including climate change adaptation, would benefit both the community and conservation efforts to adapt to issues such as increased water stress and changes in ecosystems.

Appendix 3. BP Prudhoe Bay operations

BP Exploration (Alaska) Inc. currently has operations in the arctic coastal tundra of Alaska including Prudhoe Bay, which includes one end of the Trans Alaska Pipeline System (TAPS). This arctic coastal tundra ecoregion has a climate stability of 0.079, i.e. only 8% of the ecoregion will have the same climate profile as it does now. No Environmental Impact Assessment reports could be found for these sites, although the 2013 BP Alaska Site Report (BP 2013), which consisted of a general description and environmental performance overview of its Alaska operations, and the 2014 Environmental Assessment for the North Prudhoe Bay Geophysical Seismic Survey (BP 2014), which included a description of the affected environment and impacts of 'no action' against the proposed project of this offshore operation, were explored for climate change adaptation measures.

The Alaska report states that environmental aspects are reviewed and updated annually through the EMS, suggesting scope for continual improvement and adjustment. Measures were taken for tundra protection and rehabilitation and avoidance of thermal subsidence. BP also has a wildlife monitoring programme to better understand operational effects on wildlife populations, which could also be

undertaken from an additional climate change perspective, for example to monitor changing permafrost conditions, vegetation or migratory paths.

The Seismic Survey Environmental Assessment considers climate change and potential effects on environmental conditions and natural resources in the operational area, including the effects on marine mammals and fish. The assessment appendix includes a section on climate change consideration in evaluating cumulative effects on environmental, social, and economic resources of the Arctic, particularly ocean acidification. However, what is considered is whether the operation is an additional cause of climate change or whether it is part of the cumulative impact on the environment, rather than the effect of climate change on the operation or how environmental management can actively adapt to minimise these impacts. The thirteen team members consisted primarily of biologists but also included one meteorologist who contributed information on climate change. Including personnel with expertise on climate change in arctic and marine ecosystems projects could further improve the team composition, and foster collaboration between the environmental and engineering aspects of the project.

Data Review – Framework and Sources



DataReview.xlsx

References

Acclimatise (2009). *Understanding the investment implications of adapting to climate change - oil and gas*.

AECOM (2016). *Climate change adaptation in the oil and gas industry – Final draft report*.

Agrawala, S. et al. (2011). *Private Sector Engagement in Adaptation to Climate Change: Approaches to Managing Climate Risks*, OECD Environment Working Papers, No. 39, OECD Publishing.

Bellard, C., Bertelsmeier, C., Leadley, P., Thuiller, W. and Courchamp, F. (2012). Impacts of climate change on the future of biodiversity. *Ecology Letters*, 15: 365–377.

BP (2013). *BP Exploration (Alaska) Inc. Site Report*.

BP (2014). *BP Exploration (Alaska) Inc. North Prudhoe Bay 2014 OBS Geophysical Seismic Survey Beaufort Sea, Alaska Environmental Assessment*.

Campbell, A., Kapos, V., Scharlemann, J. P.W., Bubb, P., Chenery, A., Coad, L., Dickson, B., Doswald, N., Khan, M. S. I., Kershaw, F. and Rashid, M. (2009). Review of the Literature on the Links between Biodiversity and Climate Change: Impacts, Adaptation and Mitigation. Secretariat of the Convention on Biological Diversity, Montreal. Technical Series No. 42, 124 pages.

Chapin, F. et al. (2000). Consequences of changing biodiversity. *Nature*. 405:234-242.

Cheung, W. et al. (2009). “Projecting global marine biodiversity impacts under climate change scenarios” *Fish and Fisheries* 10: 235-251.

Chevron (c2001-2016). *Managing Climate Risk*. [accessed 28 Jul 2016].

CSBI (2015). *A cross-sector guide for implementing the Mitigation Hierarchy*.

NAOC (2008). *EIA of Nimbe Field Further Development Project*.

Foden, W.B. et al. (2013). Identifying the World’s Most Climate Change Vulnerable Species: A Systematic Trait-Based Assessment of all Birds, Amphibians and Corals. *PLoS ONE* 8(6): e65427.

Hanson, C., J. Ranganathan, C. Iceland, and J. Finisdore. (2012). *The Corporate Ecosystem Services Review: Guidelines for Identifying Business Risks and Opportunities Arising from Ecosystem Change*. Version 2.0. Washington, DC: World Resources Institute.

Heller, N.E. and Zavaleta, E.S. (2009). Biodiversity management in the face of climate change: A review of 22 years of recommendations. *Biological conservation*, 142: 14-32.

ICMM (2013). *Adapting to a changing climate: implications for the mining and metals industry*.

IFC (2012). *Performance Standard 6: Biodiversity Conservation and Sustainable Management of Living Natural Resources*.

CCI, CI, WCS, CLP, and CISM (2010). *Links in biodiversity conservation, ecosystems services and climate change: teaching materials and building capacity*.

IPIECA (2013). *Addressing adaptation in the oil and gas industry*.

IPIECA, OGP (2005). *A guide to developing biodiversity action plans for the oil and gas sector*.

IPIECA, OGP (2011). *Ecosystem services guidance 8*.

Jones, M.C. and Cheung, W.W.L. (2015). Multi-model ensemble projections of climate change effects on global marine biodiversity. *ICES Journal of Marine Science*. 72(3): 741–752.

KPMG (2012). *Climate change adaptation in the private sector (UNFCCC Private Sector Initiative)*.

Pacifici, M. et al. (2015). Assessing Species Vulnerability to Climate Change. *Nature Climate Change*. 5:215–225.

PwC (2010). *Business leadership on climate change adaptation: Encouraging engagement and action*.

Segan, D.B., Murray, K.A., and Watson, J.E.M. (2016). A global assessment of current and future biodiversity vulnerability to habitat loss-climate change interactions. *Global Ecology and Conservation*. 5:12–21.

SPDC (2013). *Soku Gas Plant-San Barth Manifold Project*.

UNEP (2010). *Are you a green leader? Business and biodiversity: making the case for a lasting solution*.

UNFCCC (2011). *Assessing the costs and benefits of adaptation options: an Overview of approaches*.

UN Global Compact and UNEP (2012). *Business and climate change adaptation: Toward resilient companies and communities*.

UN Global Compact, UNFCCC, and UNEP (2015). *The business case for responsible corporate adaptation: Strengthening private sector and community resilience (A Caring for Climate report)*.

Urban, M. (2015). Accelerating extinction risk from climate change. *Science*. 348:571–573

Watson, J.E.M., Iwamura, T., and Butt, N. (2013). Mapping vulnerability and conservation adaptation strategies under climate change. *Nature Climate Change*, 3: 989–994.

Wicander, S., Helfgott, A., Bailey, M., Munroe, R., Ampomah, G., Diouf, A., Devisscher, T. and Corrigan, C. (2016). *Resilience and adaptation planning for communities in protected areas. A step-by-step guide*. Cambridge: UNEP-WCMC.

Young, B.E., Byers, E., Hammerson, G., Frances, A., Oliver, L., and Treher, A. (2015). *Guidelines for using the NatureServe Climate Change Vulnerability Index Release 3.0*. Arlington, VA: NatureServe.